

Probability

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **interpret and write** notation for compound and conditional probabilities.

RELATED SKILLS

- I Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.

☐ 2. I can **organize** categorical data in a contingency table or Venn diagram.

RELATED SKILLS

- I Construct and interpret a contingency table.
- I Represent categorical events and their overlap with a Venn diagram.

☐ 3. I can **calculate and interpret** empirical, marginal, joint, and conditional probabilities from a contingency table.

RELATED SKILLS

- I Calculate and interpret a conditional probability.
- I Calculate and interpret marginal and joint probabilities.
- I Calculate probability from observed relative frequency.

☐ 4. I can **identify** an experiment, outcomes, events, and a sample space.

RELATED SKILLS

- I Identify outcomes, events, and the sample space of an experiment.
- A Read and write probability notation and correctly use the terms outcome, event, sample space, complement, intersection, union, and conditional probability.

☐ 5. I can **calculate** theoretical probabilities when outcomes are equally likely.

RELATED SKILLS

- I Calculate theoretical probability for equally likely outcomes.
- A Identify outcomes, events, and the sample space of an experiment.

☐ 6. I can **use** complements to calculate probabilities.

RELATED SKILLS

- I Use the complement rule to calculate probability.

☐ **7. I can **calculate** probabilities of compound 'and' events and distinguish independent from dependent events.**

RELATED SKILLS

- ☐ I Calculate the probability of a compound and event.
 - ☐ I Determine whether events are independent or dependent.
 - ☐ D Calculate and interpret a conditional probability.
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☐ **8. I can **calculate** probabilities of compound 'or' events and distinguish overlapping from disjoint events.**

RELATED SKILLS

- ☐ I Calculate the probability of a compound or event.
 - ☐ I Determine whether events are overlapping or disjoint.
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☐ **9. I can **compare** experimental and theoretical probability and explain the law of large numbers.**

RELATED SKILLS

- ☐ I Explain how experimental probability stabilizes with many trials.
 - ☐ D Calculate probability from observed relative frequency.
 - ☐ D Calculate theoretical probability for equally likely outcomes.
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